

Notes: Bohren, Imas & Rosenberg (AER, 2019)

The Dynamics of Discrimination: Theory and Evidence

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to static discrimination tests	3
2.2	Contribution relative to statistical-discrimination models	3
2.3	Contribution relative to taste-based discrimination	4
2.4	Contribution relative to social-learning and reputation research	4
2.5	Contribution relative to field experiments on online platforms	4
3	Theoretical framework	4
3.1	Environment and evaluator problem	4
3.2	Discrimination	5
3.3	Initial discrimination and subjectivity	5
3.4	Dynamic discrimination with correctly specified beliefs	6
3.5	Dynamic reversal with misspecified beliefs	6
4	Experiment and data	6
4.1	Online mathematics platform	6
4.2	Gender manipulation	6
4.3	Novice and advanced accounts	7
4.4	Questions versus answers	7
4.5	Outcome variables	7
5	Empirical specifications	7
5.1	Subjectivity test	7
5.2	Dynamics test	8
5.3	Identification	8
6	Results and interpretation	8
6.1	Subjectivity results	8
6.2	Dynamic reversal results	8
7	Tables and figures	9
7.1	Figure 1: theory of reversal	9
7.2	Table 1 and Figure 2: subjectivity test	9
7.3	Figures 3 and 4: dynamic reversal in questions	10
7.4	Table 2: regression evidence for dynamics	10
8	Short summary	11
9	Conclusion	11
9.1	Core conclusion	11
9.2	Economic intuition	11

9.3	Takeaway	11
-----	--------------------	----

1 Big picture

- **Question.** Can the time path of discrimination reveal its underlying source?
- **Static identification problem.** In a one-shot evaluation, different mechanisms can generate the same observed treatment gap. A lower evaluation of women relative to men can arise from biased beliefs, rational statistical discrimination, taste-based discrimination, or preference-based partiality.
- **Dynamic insight.** When workers accumulate public evaluation histories, discrimination evolves differently depending on whether it is driven by beliefs or preferences. The paper shows that the *dynamics* of evaluation gaps identify mechanisms that static gaps cannot.
- **Core theoretical result.** Subjectivity in evaluation separates belief-based from preference-based partiality. When signal precision rises, belief-based discrimination falls because evaluators rely less on priors. Preference-based discrimination does not disappear even when the signal is precise.
- **Core dynamic result.** Under correctly specified belief-based partiality, discrimination can be mitigated as histories accumulate, but it cannot reverse sign. A dynamic reversal implies some form of misspecification in beliefs or in evaluators' models of other evaluators.
- **Experiment.** The authors run a field experiment on a large online mathematics Q&A platform. They post novel mathematics questions and answers under accounts with male or female usernames and with either no prior history or high reputation histories.
- **Main empirical findings.** For objective answer evaluations, there is little gender discrimination. For subjective question evaluations, novice female accounts are penalized. At high reputation levels, the direction reverses: advanced female accounts are favored relative to advanced male accounts.
- **Interpretation.** The combination of subjectivity effects and dynamic reversal points to biased belief-based discrimination, not preference-based discrimination and not correctly specified statistical discrimination.
- **Takeaway.** Discrimination should be studied as a dynamic process. Evaluation histories do not merely reduce discrimination; they can reverse it when observers understand that prior evaluators imposed different standards on different groups.

2 Incremental contribution of the paper

2.1 Contribution relative to static discrimination tests

- Much empirical discrimination research compares outcomes between groups in a single round: interview callbacks, hiring decisions, grades, votes, promotions, or prices.
- Static experiments can show whether discrimination exists, but they generally cannot distinguish taste-based discrimination from belief-based discrimination.
- Bohren, Imas, and Rosenberg add a dynamic identification strategy: compare discrimination at different points in an evaluation history.

Interpretation: The paper's first incremental contribution is methodological. It shows that the evolution of discrimination across histories contains information about the mechanism, not just about the size of the gap.

2.2 Contribution relative to statistical-discrimination models

- Classic statistical-discrimination models emphasize correct group-level beliefs and imperfect signals.
- In such models, better individual information should mitigate discrimination.

- The paper proves a stronger dynamic result: if evaluators correctly understand the evaluation process, a reversal of discrimination cannot occur.
- Therefore, observing a reversal identifies biased or misspecified belief-based partiality.

Interpretation: This is more than a test for statistical discrimination. It distinguishes correctly specified belief-based discrimination from biased belief-based discrimination.

2.3 Contribution relative to taste-based discrimination

- Taste-based or preference-based discrimination can persist even when quality is perfectly observable.
- The paper uses the distinction between answers and questions to vary subjectivity. Answers are relatively objective because correctness is easier to assess; questions are more subjective because quality depends on interest, novelty, importance, and research effort.
- The finding that gender gaps arise for questions but not answers is consistent with belief-based partiality and inconsistent with a pure animus/taste story.

Interpretation: The paper turns subjectivity into a diagnostic tool. More subjective evaluations provide more room for priors and stereotypes to affect judgments.

2.4 Contribution relative to social-learning and reputation research

- Reputation histories are usually treated as informative signals that reduce uncertainty.
- This paper emphasizes that histories also encode evaluators' prior standards.
- If women must meet a higher standard to receive the same evaluation, a given positive history can be more informative for women than for men.
- Evaluators who understand this can update enough to reverse their rankings of male and female accounts with the same observed reputations.

Interpretation: The paper's dynamic contribution is that reputations are endogenous social signals. The meaning of a reputation depends on how earlier evaluators treated the group.

2.5 Contribution relative to field experiments on online platforms

- The field setting allows the authors to manipulate usernames and account histories while holding post content fixed through random assignment.
- The experiment separately manipulates gender, reputation history, and post type.
- This design connects the theoretical mechanisms directly to observable user behavior on a real platform.

Interpretation: The design is unusually clean for testing discrimination dynamics because accounts can be assigned histories and names while the posted content is randomized.

3 Theoretical framework

3.1 Environment and evaluator problem

There are workers who repeatedly produce tasks. A worker has gender $g \in \{F, M\}$, ability a , and task quality q . Evaluators observe a history h of prior evaluations and a noisy signal s of current task quality. Each evaluator chooses an evaluation v .

An evaluator of type i chooses v to minimize expected squared loss around perceived quality net of preference costs:

$$v_i(h, s, g) = \arg \max_{v \in \mathbb{R}} \widehat{E}_i \left[- (v - (q - c_g^i))^2 \mid h, s, g \right]. \quad (1)$$

The optimal evaluation is

$$v_i(h, s, g) = \widehat{E}_i[q \mid h, s, g] - c_g^i. \quad (2)$$

- $\widehat{E}_i$ is expectation under the evaluator's subjective model.
- c_g^i captures preference-based partiality or animus toward group g .
- Belief-based partiality appears through different priors about average ability by gender.

Interpretation: The model separates beliefs from preferences. Evaluators can treat women differently because they expect lower quality or because they dislike rewarding women.

3.2 Discrimination

For a fixed history and signal, type i 's discrimination is the evaluation gap:

$$D_i(h, s) = v_i(h, s, M) - v_i(h, s, F). \quad (3)$$

Aggregate discrimination is

$$D(h, s) = E_\pi[D_i(h, s)]. \quad (4)$$

A woman faces discrimination if $D_i(h, s) > 0$ or $D(h, s) > 0$. A man faces discrimination if $D_i(h, s) < 0$ or $D(h, s) < 0$.

Interpretation: Discrimination is a behavioral object: it is measured by evaluations. Partiality is a primitive: it comes from beliefs and preferences.

3.3 Initial discrimination and subjectivity

In the first period, the evaluator's posterior expected quality is a weighted average of the prior mean and the signal:

$$v(h_1, s_1, g) = \frac{\tau_q \widehat{\mu}_g + \tau_\eta s_1}{\tau_q + \tau_\eta} - c_g. \quad (5)$$

Initial discrimination is

$$D(h_1, s_1) = \frac{\tau_q}{\tau_q + \tau_\eta} (\widehat{\mu}_M - \widehat{\mu}_F) + c_F. \quad (6)$$

- τ_η is signal precision.
- Higher signal precision makes the signal more objective.
- Belief-based discrimination declines as τ_η increases.
- Preference-based discrimination remains when τ_η increases.

Interpretation: This delivers Proposition 1. Subjective judgments create room for stereotypes or biased priors. Objective judgments leave less room for belief-based partiality.

3.4 Dynamic discrimination with correctly specified beliefs

If all evaluators share a common prior and correctly understand how others evaluate, a given evaluation is more informative for a group held to a higher standard. Thus, discrimination falls over time as histories accumulate.

However, the paper proves that this mitigation cannot become a reversal. For a fixed evaluation history, the posterior belief about male ability remains above the posterior belief about female ability if it started that way and the model is correctly specified.

Interpretation: Correct statistical discrimination can fade, but it cannot flip direction. Dynamic reversal is therefore a strong diagnostic.

3.5 Dynamic reversal with misspecified beliefs

The paper then considers a setting with two evaluator types:

- a heuristic type who believes men are more able and incorrectly thinks all evaluators share this belief;
- an impartial type who has no gender prior bias but understands that some evaluators impose higher standards on women.

The impartial type may infer that a female account with the same history as a male account is actually higher quality because the female account had to overcome biased prior evaluators. If enough impartial evaluators recognize this, aggregate discrimination can reverse.

Interpretation: Reversal is evidence of biased belief-based partiality with misspecified social learning. It is not predicted by simple taste discrimination or correctly specified statistical discrimination.

4 Experiment and data

4.1 Online mathematics platform

- The field experiment is conducted on a large online mathematics Q&A forum.
- Users post questions and answers, which other users evaluate using upvotes and downvotes.
- The platform’s reputation system converts votes into reputation points.
- Usernames and reputation histories are visible, making the site a natural environment for studying repeated evaluation.

Interpretation: The platform has the key elements of the theory: repeated tasks, public histories, noisy evaluation, and visible group cues.

4.2 Gender manipulation

- The authors create accounts with male and female first names.
- Names are drawn from common names from the 2000s.
- Gender is manipulated through the username while content is randomly assigned to accounts.
- This holds post quality fixed in expectation across male and female accounts.

Interpretation: The design isolates differential evaluation due to perceived gender rather than content quality.

4.3 Novice and advanced accounts

- Novice accounts have no prior posts or reputation history.
- Advanced accounts have high reputation histories.
- The authors reassign gendered usernames to advanced accounts so that high-reputation male and female accounts have similar reputation levels.
- Novel questions are randomly assigned to male/female and novice/advanced conditions.

Interpretation: This manipulation creates exogenous evaluation histories. It is the empirical counterpart of comparing discrimination at different histories h .

4.4 Questions versus answers

- Questions are more subjective to evaluate: they may be interesting, novel, important, well-written, or useful.
- Answers are more objective because correctness is easier to assess.
- The authors post 280 novel mathematics questions and 140 original answers.
- Outcomes are measured seven days after posting.

Interpretation: The question-answer comparison tests whether discrimination rises with subjectivity, as predicted by belief-based discrimination.

4.5 Outcome variables

The paper uses two main outcomes:

$$\Delta Rep = \text{change in account reputation from the post,} \quad (7)$$

$$NetVotes = \text{upvotes minus downvotes on the post.} \quad (8)$$

Interpretation: Reputation points and net votes are the platform's real evaluative outcomes. They map directly into the model's evaluation variable.

5 Empirical specifications

5.1 Subjectivity test

For novice posts, the authors estimate regressions of the form:

$$Y_i = \alpha + \beta_1 Male_i + \beta_2 Question_i + \beta_3 (Male_i \times Question_i) + \varepsilon_i. \quad (9)$$

- Y_i is either ΔRep or net votes.
- $Male_i$ equals one for male username accounts.
- $Question_i$ equals one for questions and zero for answers.
- β_3 tests whether gender discrimination differs between subjective questions and more objective answers.

Interpretation: A positive male advantage on questions but not answers supports belief-based partiality.

5.2 Dynamics test

For questions, the authors compare novice and advanced accounts:

$$Y_i = \alpha + \gamma_1 \text{Male}_i + \gamma_2 \text{Advanced}_i + \gamma_3 (\text{Male}_i \times \text{Advanced}_i) + \varepsilon_i. \quad (10)$$

- Advanced_i equals one for high-reputation accounts.
- γ_1 measures the male advantage among novice accounts.
- $\gamma_1 + \gamma_3$ measures the male advantage among advanced accounts.
- A negative γ_3 large enough to overturn γ_1 indicates dynamic reversal.

Interpretation: The dynamics regression tests whether discrimination merely decreases or actually reverses when reputation history is positive.

5.3 Identification

- Gender is assigned by account username.
- Questions and answers are randomly assigned to accounts.
- Advanced account gender is randomized by renaming accounts, so male and female advanced accounts have comparable histories.
- Research assistants writing answers do not know which gendered account will post the answer.
- The posting schedule is predetermined to avoid strategic timing or flooding the platform.

Interpretation: Random assignment of gender labels and content makes the evaluation differences interpretable as discrimination in the platform’s response.

6 Results and interpretation

6.1 Subjectivity results

- Answers posted by male and female novice accounts receive statistically similar evaluations.
- Questions posted by male novice accounts receive more reputation points and net votes than questions posted by female novice accounts.
- The male-by-question interaction in Table 1 is positive and significant for both reputation and net votes.

Interpretation: Discrimination appears when evaluation is more subjective. This supports belief-based partiality rather than simple preference-based animus.

6.2 Dynamic reversal results

- Novice female accounts are penalized relative to novice male accounts for questions.
- Advanced female accounts are favored relative to advanced male accounts for questions.
- The interaction between male and advanced account status is negative and statistically significant.
- The reversal is visible in both average reputation changes and the CDF of reputation changes.

Interpretation: The reversal implies that positive histories are interpreted differently by gender. Evaluators infer that women with high reputation likely overcame stricter earlier standards.

7 Tables and figures

7.1 Figure 1: theory of reversal

Read:

- Figure 1 plots second-period discrimination as a function of the share of heuristic evaluators.
- For some parameter values, discrimination against females in the first period becomes discrimination against males in the second period.
- The reversal occurs when impartial evaluators understand that biased evaluators imposed higher standards on women.

Economic message: The figure illustrates the main theoretical insight: a reversal is possible under biased belief-based partiality with misspecified social learning.

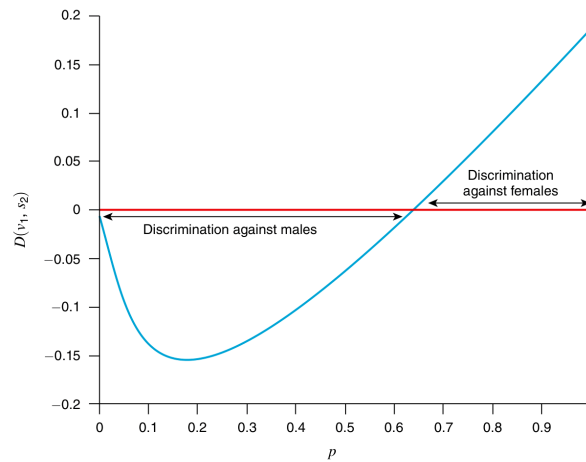


FIGURE 1. SECOND PERIOD DISCRIMINATION, AS A FUNCTION OF THE PROPORTION OF HEURISTIC EVALUATORS

Proposition 3 does not rely on the assumption that the impartial evaluators exactly understand the bias of the heuristic evaluators or accurately estimate their prevalence in the population. It is straightforward to derive a similar result when the impartial evaluators under- or overestimate either the level of the bias of heuristic

7.2 Table 1 and Figure 2: subjectivity test

Read:

- Table 1 shows no significant gender gap for answers, which are relatively objective.
- For questions, the male coefficient is positive: male novice accounts receive more reputation points and more net votes.
- The interaction between male and question is positive and significant.
- Figure 2 shows the same answer result visually: average reputation changes and the distribution of reputation changes are similar for male and female answer accounts.

Economic message: Gender discrimination appears in subjective evaluations of questions, but not in more objective evaluations of answers. This pattern supports belief-based rather than preference-based discrimination.

TABLE 1—SUBJECTIVITY: EFFECT OF GENDER ON EVALUATION OF NOVICE ANSWERS AND QUESTIONS

	Answers only		Questions only		Answers and questions	
	ΔRep (1)	Net votes (2)	ΔRep (3)	Net votes (4)	ΔRep (5)	Net votes (6)
Male	-1.38 (0.97)	-0.31 (0.17)	2.86 (1.32)	0.58 (0.27)	-1.38 (1.16)	-0.31 (0.22)
Question					0.08 (1.16)	0.09 (0.22)
Male $\times$ question					4.24 (1.64)	0.89 (0.32)
Constant	4.60 (0.69)	0.79 (0.12)	4.68 (0.93)	0.88 (0.19)	4.60 (0.82)	0.79 (0.16)
Observations	135	135	135	135	270	270

Notes: Standard errors from OLS regressions reported in parentheses; *Male* = 1 if male username, 0 otherwise; *Question* = 1 if question post, 0 if answer; Novice accounts only.

Table 1: subjectivity test

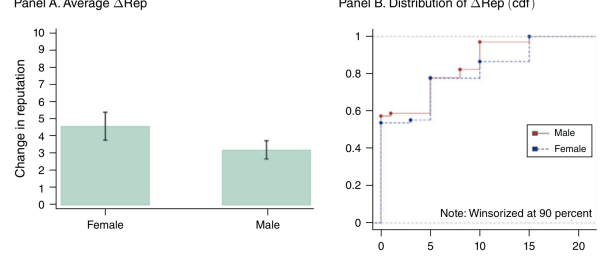


FIGURE 2. CHANGE IN REPUTATION FOR ANSWERS

Figure 2: answers, average and CDF of reputation changes

7.3 Figures 3 and 4: dynamic reversal in questions

Read:

- Figure 3 shows average reputation changes for questions by gender and reputation status.
- Among novice accounts, male accounts do better than female accounts.
- Among advanced accounts, female accounts do better than male accounts.
- Figure 4 shows the same reversal in the CDFs of reputation changes.

Economic message: The visual evidence shows dynamic reversal: gender discrimination changes sign as reputation history improves.

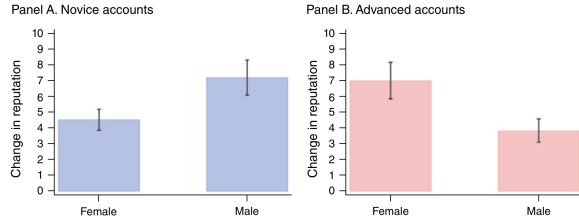


FIGURE 3. AVERAGE CHANGE IN REPUTATION FOR QUESTIONS

Figure 3: questions, average reputation changes

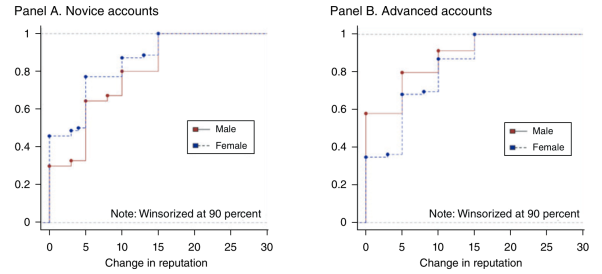


FIGURE 4. DISTRIBUTION OF CHANGE IN REPUTATION FOR QUESTIONS (CDF)

Figure 4: questions, CDFs of reputation changes

7.4 Table 2: regression evidence for dynamics

Read:

- Table 2 estimates the gender effect for advanced accounts and for combined novice/advanced samples.
- Among advanced accounts, the male coefficient is negative for both reputation and net votes.
- In the combined sample, the male-by-advanced interaction is negative and significant.
- The binary-upvote specification confirms that the reversal is not driven only by extreme outliers.

Economic message: The regression evidence formally confirms the dynamic reversal. This is the key empirical fact used to identify biased belief-based discrimination.

TABLE 2—DYNAMICS: EFFECT OF GENDER ON EVALUATION QUESTIONS, NOVICE AND ADVANCED

	Advanced		Novice and advanced		
	ΔRep	Net votes	ΔRep	Net votes	Binary
	(1)	(2)	(3)	(4)	(5)
Male	-3.16 (1.37)	-0.62 (0.28)	2.86 (1.36)	0.58 (0.27)	0.17 (0.08)
Advanced			2.33 (1.35)	0.49 (0.27)	0.09 (0.08)
Male $\times$ advanced			-6.02 (1.91)	-1.20 (0.38)	-0.40 (0.11)
Constant	7.01 (0.97)	1.38 (0.20)	4.68 (0.96)	0.88 (0.19)	0.56 (0.06)
Observations	138	138	273	273	273

Notes: Standard errors from OLS regressions reported in parentheses; *Male* = 1 if male username, 0 otherwise; *Advanced* = 1 if Advanced account, 0 otherwise.

8 Short summary

- The paper studies how discrimination evolves with evaluation histories.
- Static discrimination cannot identify whether the source is preference-based or belief-based.
- The theory shows that signal precision and history dynamics identify different sources.
- Belief-based discrimination falls when evaluation becomes more objective.
- Preference-based discrimination remains even under objective evaluation.
- Correctly specified belief-based discrimination can be mitigated by history but cannot reverse.
- A reversal indicates biased or misspecified belief-based partiality.
- The field experiment manipulates gendered usernames and account reputation on an online mathematics forum.
- Novice female questions are penalized, but advanced female questions are favored.
- Answers, which are more objective, show little gender discrimination.
- The evidence supports biased belief-based discrimination rather than taste-based discrimination.

9 Conclusion

9.1 Core conclusion

Bohren, Imas, and Rosenberg show that discrimination is dynamic and that its time path can reveal its underlying source. A static evaluation gap is insufficient to distinguish preferences from beliefs. But a subjectivity comparison and a dynamic reversal provide strong evidence of biased belief-based discrimination.

9.2 Economic intuition

If women face stricter initial standards, then a positive history can be more informative for a woman than for a man. Evaluators who understand this may eventually infer that a high-reputation woman is stronger than a high-reputation man with the same observed history. This creates a reversal.

9.3 Takeaway

The paper’s central lesson is that reputation systems can both transmit and reveal discrimination. Histories mitigate some discrimination, but when histories are interpreted through awareness of prior bias, they can reverse discrimination. Empirically, this means researchers should study not only whether disadvantaged groups are penalized at entry, but how evaluation gaps evolve as evidence accumulates.